

MATH Semester Examination 2025

Mathematics (MATH) · Semester Examination · 2025

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Compare Rank of a Matrix with Systems of Linear Equations. When would you prefer one over the other? (5 marks)
2. Discuss the role of Mean Value Theorems in Differential Calculus. (5 marks)
3. Illustrate Conditional Probability with a worked example. (5 marks)
4. Write short notes on Propositional Logic. (5 marks)
5. What is meant by Permutations and Combinations? Discuss its advantages and limitations. (5 marks)

Part B

1. What is meant by Rank of a Matrix? Discuss its advantages and limitations. (Unit 1)
2. Define Systems of Linear Equations. Derive or justify the key properties involved. (Unit 1)
3. Discuss the role of Mean Value Theorems in Differential Calculus. (Unit 2)
4. Write short notes on Taylor and Maclaurin Series. (Unit 2)
5. What is meant by Conditional Probability? Discuss its advantages and limitations. (Unit 3)
6. What is meant by Bayes' Theorem? Discuss its advantages and limitations. (Unit 3)
7. Illustrate Propositional Logic with a worked example. (Unit 4)
8. Compare Predicate Logic with Mathematical Induction. When would you prefer one over the other? (Unit 4)
9. What is meant by Permutations and Combinations? Discuss its advantages and limitations. (Unit 5)
10. Illustrate Pigeonhole Principle with a worked example. (Unit 5)